

Compliance Test Harness Quickstart

For

A-GRA

5.0a

30 APR 2026

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Revision History

Version	Description	Generation Date
5.0a	Initial Public ASK 5.0a Release	30 APR 2026

Overview

This quick start guide provides instructions for using the Test Harness. This guide walks through using the Test Harness helper scripts and an example testing the Test Harness against itself for the **A-GRA 5.0a ICD**.

Follow the steps below to perform testing with the Test Harness:

1. **Install the Test Harness & ActiveMQ.**
2. **Start an ActiveMQ broker for both services.**
3. **Run one of the example scripts to either test the Test Harness against itself (shown in this example) or a System Under Test (SUT).**
4. **Review the generated test report(s).**

Prerequisites

Before beginning this guide, ensure you have completed the prerequisites for the test harness:

- **Test Harness:** See Test Harness System Requirements and Prerequisites.pdf

Both sets of prerequisites must be completed before proceeding with this quick start guide.

Setup

Installing the Test Harness

The Test Harness is provided as a standalone ZIP file `test-harness-ASK_5.0a_v1.2.2.zip`. To use the Test Harness, extract the contents of the ZIP file before proceeding with installation or configuration.

```
$ unzip -q test-harness-ASK_5.0a_v1.2.2.zip
$ ls -l
README.md
compliance-test-harness-1.2.2.a.tar
amq-6.1.2.tar
common_utils
compliance_test_harness
compose
configs
docs
release-notes.md
test-harness-ASK_5.0a_v1.2.2.zip
test-harness-runner.sh
test-harness-test.sh
```

To install the Test Harness image to your local podman image registry you can then run the following command:

```
$ ./scripts/setup --podman
Getting image source signatures
Copying blob 931b7ff0cb6f skipped: already exists
Copying blob 6b55d85ee844 skipped: already exists
Copying blob b4bca94bcd8f8 skipped: already exists
Copying blob 783ce3350b0f skipped: already exists
Copying blob 52e87c25af26 skipped: already exists
Copying blob 463c8eb77555 skipped: already exists
Copying blob 6b442d70702d skipped: already exists
Copying blob 4a20195fd08e skipped: already exists
Copying config 09c653440e done |
Writing manifest to image destination
Loaded image: docker.io/apache/activemq-classic:6.1.2
Getting image source signatures
Copying blob f63eb04151bc skipped: already exists
Copying blob c1d21f2ad04b skipped: already exists
Copying blob ddd9beabadf5 skipped: already exists
Copying blob 4f4fb700ef54 skipped: already exists
Copying blob 5c57c92aelde skipped: already exists
Copying blob 915db0c951db skipped: already exists
Copying config e61837b577 done |
Writing manifest to image destination
Loaded image: docker.io/library/compliance-test-harness:1.2.2.a
```

If you wish to also install only the Test Harness image you can run `./scripts/setup-test-harness --podman` instead.

The Test Harness image should now be installed to your podman image registry. Verify the installation by running the help command:

```
$ podman image ls | grep test-harness
docker.io/library/compliance-test-harness
1.2.2a                                0dee0322d0a1  16 hours ago  1.34 GB
```

You can similarly check that the activemq image is installed:

```
podman image ls | grep activemq-classic
docker.io/apache/activemq-classic
6.1.2                                09c653440eb3  21 months ago  387 MB
```

A bash script (`test-harness-runner.sh`) has been provided for streamlining basic test harness workflows. The example further in this guide will show how to use this script to do basic interface suite testing.

```
$ ./test-harness-runner.sh -h
Usage: ./test-harness-runner.sh [OPTIONS]
```

Run test harness sequences for a specific system under test.

Options:

```

    -h, --help                Show this help message
    -p, --stomp-port <port>  STOMP port for ActiveMQ host networking
                              (default: 61613)

    Note: test_harness connects to this port if
    using host networking

    --openwire-port <port>   OpenWire port for ActiveMQ (default: 61616)
    --sut <system>           System under test (default: all)

    MA ENDPOINT TESTING (test-harness simulates
    external system → MA):
    MA, C2 → MA, MA → MA)

    ma          - All MA Interfaces (VI → MA, MS →
    ma-vi        - Vehicle Interface - VI (VI → MA)
    ma-ms        - Mission Systems - MS (MS → MA)
    ma-c2        - Command and Control - C2 (C2 →
    MA)

    p2p          - ACP 1 MA - MA (MA → MA)

    EXTERNAL ENDPOINT TESTING (test-harness
    simulates MA → external system):
    vi          - MA - VI (MA → VI)
    ms          - MA - MS (MA → MS)
    c2          - MA - C2 (MA → C2)

    --sequence <sequence>    Specific sequence to test for the specified
    system under test (Optional: requires sut)
    --custom <sequences>      List of sequence pytests to run (Optional:
    ignores sut)
    -l, --list                Prints list of sequences for a given system
    under test (Optional)
    -n, --network <network>   Configures the ActiveMQ podman network
    (Optional: requires alias)
    -a, --network-alias <alias> Configures the name of the ActiveMQ podman
    container (Optional: requires network)
    --output <dir>            Output directory for results (default: ./test-
    harness-logs)

    Examples:
    ./test-harness-runner.sh --sut vi --stomp-port 5152
    # Custom STOMP port
    ./test-harness-runner.sh --sut ma-ms --output /tmp/test-results
    # Custom output dir
    ./test-harness-runner.sh --sut c2
    ./test-harness-runner.sh --sut vi --stomp-port 45455 --openwire-port 45454
    # Custom ports
    ./test-harness-runner.sh --sut p2p --stomp-port 9999
    # Custom STOMP port
    ./test-harness-runner.sh -h

    Notes:
    - Standard ActiveMQ ports: STOMP=61613

```

The Test-Harness Environment

Test-harness expects to use an already existing environment. It is assumed the user has already prepared an environment for testing that includes their system under test and ActiveMQ.

Running the Test Harness

The Test Harness supports multiple execution methods. One common approach uses a shared ActiveMQ broker that both your Autonomy services and the Test Harness connect to. With this method, launch your services on the message bus, then start the Test Harness to validate the sequences.

Launching ActiveMQ

Start the bundled ActiveMQ container. Remember the STOMP port used by the ActiveMQ container (default: 61613) — you will need this port number later when connecting to the Test Harness (you can optionally specify a different port in `compose/asb-compose.yml` if there are multiple users on your machine).

```
$ ./scripts/run-asb --podman
podman-compose version: 1.0.6
['podman', '--version', '']
using podman version: 4.9.3
** excluding: set()
podman stop -t 10 amq-asb
Error: no container with name or ID "amq-asb" found: no such container
exit code: 125
podman rm amq-asb
Error: no container with ID or name "amq-asb" found: no such container
exit code: 1
podman-compose version: 1.0.6
['podman', '--version', '']
using podman version: 4.9.3
** excluding: set()
['podman', 'ps', '--filter', 'label=io.podman.compose.project=compose', '-a',
'--format', '{{ index .Labels "io.podman.compose.config-hash"}}']
['podman', 'network', 'exists', 'compose_default']
podman create --name=amq-asb --label io.podman.compose.config-
hash=eece4d37b87f275af8f0ca54e1dab049c74c3d8583dba0f28793af94e263ef9a --label
io.podman.compose.project=compose --label io.podman.compose.version=1.0.6 --
label PODMAN_SYSTEMD_UNIT=podman-compose@compose.service --label
com.docker.compose.project=compose --label
com.docker.compose.project.working_dir=/home/<user>/case/test-harness-
mk3/compose --label com.docker.compose.project.config_files=./compose/asb-
compose.yml --label com.docker.compose.container-number=1 --label
com.docker.compose.service=activemq --net compose_default --network-alias
activemq -p 8161:8161 -p 61616:61616 -p 61613:61613 --healthcheck-command
/bin/sh -c 'echo ok' --healthcheck-interval 10s --healthcheck-timeout 3s --
healthcheck-start-period 3s --healthcheck-retries 1 apache/activemq-
classic:6.1.2
8d38a67a1c5ce491c1f644d469bff02ab430b832d369f04238079a9843ff8680
exit code: 0
podman start -a amq-asb
[activemq] | INFO: Loading '/opt/apache-activemq/bin/setenv'
[activemq] | INFO: Using java '/opt/java/openjdk/bin/java'
[activemq] | INFO: Starting in foreground, this is just for debugging
purposes (stop process by pressing CTRL+C)
```

```

[activemq] | INFO: Creating pidfile /opt/apache-activemq/data/activemq.pid
[activemq] | Java Runtime: Eclipse Adoptium 17.0.11 /opt/java/openjdk
[activemq] |   Heap sizes: current=67584k free=61108k max=1048576k
[activemq] |   JVM args: -Xms64M -Xmx1G -
Djava.util.logging.config.file=logging.properties -
Djava.security.auth.login.config=/opt/apache-activemq/conf/login.config -
Djetty.host=0.0.0.0 -Dcom.sun.management.jmxremote -Djava.awt.headless=true -
Djava.io.tmpdir=/opt/apache-activemq/tmp --add-reads=java.xml=java.logging --
add-opens=java.base/java.security=ALL-UNNAMED --add-
opens=java.base/java.net=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-
UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-
opens=java.naming/javax.naming.spi=ALL-UNNAMED --add-
opens=java.rmi/sun.rmi.transport.tcp=ALL-UNNAMED --add-
opens=java.base/java.util.concurrent=ALL-UNNAMED --add-
opens=java.base/java.util.concurrent.atomic=ALL-UNNAMED --add-
opens=java.base/sun.nio.ch=ALL-UNNAMED --add-
exports=java.base/sun.net.www.protocol.http=ALL-UNNAMED --add-
exports=java.base/sun.net.www.protocol.https=ALL-UNNAMED --add-
exports=java.base/sun.net.www.protocol.jar=ALL-UNNAMED --add-
exports=jdk.xml.dom/org.w3c.dom.html=ALL-UNNAMED --add-
exports=jdk.naming.rmi/com.sun.jndi.url.rmi=ALL-UNNAMED -
Dactivemq.classpath=/opt/apache-activemq/conf:/opt/apache-activemq/./lib/: -
Dactivemq.home=/opt/apache-activemq -Dactivemq.base=/opt/apache-activemq -
Dactivemq.conf=/opt/apache-activemq/conf -Dactivemq.data=/opt/apache-
activemq/data -Djolokia.conf=file:/opt/apache-activemq/conf/jolokia-
access.xml
[activemq] | Extensions classpath:
[activemq] |   [/opt/apache-activemq/lib,/opt/apache-
activemq/lib/camel,/opt/apache-activemq/lib/optional,/opt/apache-
activemq/lib/web,/opt/apache-activemq/lib/extra]
[activemq] | ACTIVEMQ_HOME: /opt/apache-activemq
[activemq] | ACTIVEMQ_BASE: /opt/apache-activemq
[activemq] | ACTIVEMQ_CONF: /opt/apache-activemq/conf
[activemq] | ACTIVEMQ_DATA: /opt/apache-activemq/data
[activemq] | Loading message broker from: xbean:activemq.xml
[activemq] | INFO | Using Persistence Adapter:
KahaDBPersistenceAdapter[/opt/apache-activemq/data/kahadb]
[activemq] | INFO | Starting Persistence Adapter:
KahaDBPersistenceAdapter[/opt/apache-activemq/data/kahadb]
[activemq] | INFO | Starting KahaDBStore
[activemq] | INFO | Opening MessageDatabase
[activemq] | INFO | Starting Temp Data Store
[activemq] | INFO | PListStore:[/opt/apache-
activemq/data/localhost/tmp_storage] started
[activemq] | INFO | Starting Job Scheduler Store
[activemq] | INFO | Persistence Adapter successfully started
[activemq] | INFO | Apache ActiveMQ 6.1.2 (localhost, ID:8d38a67a1c5c-39929-
1776706621554-0:1) is starting
[activemq] | INFO | Listening for connections at:
tcp://8d38a67a1c5c:61616?maximumConnections=1000&wireFormat.maxFrameSize=1048
57600
[activemq] | INFO | Connector openwire started
[activemq] | INFO | Listening for connections at:
amqp://8d38a67a1c5c:5672?maximumConnections=1000&wireFormat.maxFrameSize=1048
57600
[activemq] | INFO | Connector amqp started

```

```

[activemq] | INFO | Listening for connections at:
stomp://8d38a67a1c5c:61613?maximumConnections=1000&wireFormat.maxFrameSize=10
4857600
[activemq] | INFO | Connector stomp started
[activemq] | INFO | Listening for connections at:
mqtt://8d38a67a1c5c:1883?maximumConnections=1000&wireFormat.maxFrameSize=1048
57600
[activemq] | INFO | Connector mqtt started
[activemq] | INFO | Starting Jetty server
[activemq] | INFO | Creating Jetty connector
[activemq] | WARN |
ServletContext@o.e.j.s.ServletContextHandler@7a0e7ecd{/,null,STARTING} has
uncovered HTTP methods for the following paths: [/]
[activemq] | INFO | Listening for connections at
ws://8d38a67a1c5c:61614?maximumConnections=1000&wireFormat.maxFrameSize=10485
7600
[activemq] | INFO | Connector ws started
[activemq] | INFO | Apache ActiveMQ 6.1.2 (localhost, ID:8d38a67a1c5c-39929-
1776706621554-0:1) started
[activemq] | INFO | For help or more information please see:
http://activemq.apache.org
[activemq] | INFO | ActiveMQ WebConsole available at http://0.0.0.0:8161/
[activemq] | INFO | ActiveMQ Jolokia REST API available at
http://0.0.0.0:8161/api/jolokia/

```

Launching the Test Harness

Change Notice

The test suite naming scheme and the corresponding test interface directionality in this release is different from the prior test harness!

Please read this guide carefully to ensure you are selecting the correct test suite.

First, we will list the sequences that are available for test. If Vehicle Interface is the System Under Test (SUT), we specify the `--sut vi` argument to filter to just the tests applicable to testing it. Running this same command without the `--list` command will run all of the listed sequences. Add a `--sequence <sequence_name>` to run a specific sequence from this list.

```

$ ./test-harness-runner.sh --list --sut vi
activate_route
checksum_validation
collision_avoidance
control_by_curve_following
control_by_hsa_csa_command
control_by_waypoint_following
control_mode_authorization
convert_and_upload_route
exchange_heartbeat_subsystem_status_reports
intra_vehicle_comms_failure
ma_failsafe
ma_vi_command_task
mechanical_damage_reporting
modify_capabilities

```



```

prepare_for_route_activation
publish_control_status
query_airfield_update
query_for_missing_data
query_route_plan
receive_barometric_pressure
receive_control_request
receive_deactivate_route
receive_execution_status
receive_vehicle_performance_values
receive_vehicle_state_data
request_terrain_data
route_plan_data_validation
sensor_failure
unpair_control_assignment
update_c2_control_designations
validate_release_envelope
validate_route_plan
vehicle_status_reporting
vi_deactivate_route
vi_responds_to_query_for_flight_capabilities
vi_updates_to_cop

```

Each L1 component includes test items applicable to testing that L1 component (e.g., `vi`, `ms`, `c2`, `ma`) as the system under test. Additionally, a subset of the Mission Autonomy tests can be run for a specific L1 component rather than running all of the MA tests. These are indicated by the suffix `_<L1-component>` (e.g., `ma_vi`, `ma_ms`, `ma_c2`, `ma_ma`).

As this package does not contain an example L1 system to test against, we have instead provided an additional script `test-harness-test.sh` with similar arguments that starts up two instances of the test harness executing opposite sides of each sequence. This is useful for demonstration purposes and to validate the test harness's ability to pass its own tests when running both directions of an interaction.

For this test, we will run the Test Harness test script with the following arguments:

- Use the existing environment launched in the previous steps by specifying the network (`-n`) and alias (`-a`)

```

$ ./test-harness-test.sh -n ask-network -a activemq-container
=====
Test Harness Runner
=====
System Under Test:  vi
Sequence:           ALL
STOMP Port:         61613
OpenWire Port:      61616
Output:             /home/<user>/case/test-harness-mk3/test-harness-logs
=====

/app/compliance_test_harness/VI/test_activate_route.py
/app/compliance_test_harness/MA/VI/test_activate_route.py

```

```

mv: cannot stat '/home/<user>/case/test-harness-mk3/test-harness-logs/*.log':
No such file or directory
Running /app/compliance_test_harness/VI/test_activate_route.py
Using host networking
Starting up first test harness...
Starting up second test harness...
===== test session starts
=====
platform linux -- Python 3.10.12, pytest-8.4.2, pluggy-1.6.0 --
/app/.venv/bin/python
cachedir: .pytest_cache
metadata: {'Python': '3.10.12', 'Platform': 'Linux-6.6.87.2-microsoft-
standard-WSL2-x86_64-with-glibc2.35', 'Packages': {'pytest': '8.4.2',
'pluggy': '1.6.0'}, 'Plugins': {'json-report': '1.5.0', 'instafail': '0.5.0',
'metadata': '3.1.1'}}
rootdir: /app
configfile: pyproject.toml
plugins: json-report-1.5.0, instafail-0.5.0, metadata-3.1.1
collecting ...
----- live log collection -----
---
INFO      stomp.py:transport.py:786 established connection to host localhost,
port 61613
INFO      root:amq_asb_utils.py:89 Connected to ActiveMQ broker at
localhost:61613
INFO      root:amq_asb_utils.py:978 AMQ connection established on module
import
collected 1 item

compliance_test_harness/VI/test_activate_route.py::test_activate_route
----- live log setup -----
---
INFO      compliance_test_harness.conftest:conftest.py:54 Loaded
configuration from /app/test/configs/test-config.yaml successfully.
INFO      compliance_test_harness.conftest:conftest.py:55 Configuration
details: {
  "message_timeout_seconds": 6.0,
  "xml_generation_seed": 42,
  "disable_offboard_msg_wrapping": false,
  "offboard_consumer_config": {
    "wrapping_strategy": "tx"
  },
  "offboard_publisher_config": {
    "wrapping_strategy": "rx"
  },
  "platform_system_id": {
    "uuid_generation_namespace": {
      "namespace_uuid_name": "SUT",
      "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "1",
    "system_label": "1",
    "system_uuid_override": null,
    "system_uuid": "c5a0850c-282d-3211-b852-b92eb618a4ad"
  },
  "c2_system_id": {
    "uuid_generation_namespace": {

```

```

        "namespace_uuid_name": "SUT",
        "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "C2-1",
    "system_label": "C2 System",
    "system_uuid_override": null,
    "system_uuid": "a2f74db0-b0bb-33b4-b01b-744919f547bd"
},
"c2_system_id_2": {
    "uuid_generation_namespace": {
        "namespace_uuid_name": "SUT",
        "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "C2-2",
    "system_label": "C2 System 2",
    "system_uuid_override": null,
    "system_uuid": "5a64dc4d-d596-324c-88b8-7da2a0215651"
},
"peer_system_id": {
    "uuid_generation_namespace": {
        "namespace_uuid_name": "SUT",
        "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "2",
    "system_label": "Peer System",
    "system_uuid_override": null,
    "system_uuid": "31dbe31a-de46-39cc-92de-38d439ac2ab7"
}
}
}
----- live log call -----
---
INFO      root:amq_asb_utils.py:196 Subscribed to /topic/FileLocation<None>
INFO      root:amq_asb_utils.py:196 Subscribed to /topic/FileMetadata<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_MissionPlanActivationCommandStatus<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_SystemNotification<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_MissionPlanActivationCommandStatus<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_MissionPlanActivationCommandStatus<None>
INFO
compliance_test_harness.VI.test_activate_route:test_activate_route.py:118
Sending MA_MissionPlanActivationCommand
INFO      common_utils.amq_asb_utils:amq_asb_utils.py:736 Publishing message
to VI without offboard wrapping
INFO      root:amq_asb_utils.py:770 Published message to
/topic/MA_MissionPlanActivationCommand<None>
FAILED
[100%]
/app/compliance_test_harness/VI/test_activate_route.py:146: AssertionError:
No MA_MissionPlanActivationCommandStatus message received within 15.0 seconds

===== short test summary info
=====

```

```

FAILED compliance_test_harness/VI/test_activate_route.py::test_activate_route
- AssertionError: No MA_MissionPlanActivationCommandStatus message received
within 15.0 seconds
assert None
===== 1 failed in 22.81s
=====
Running /app/compliance_test_harness/VI/test_activate_route.py in reverse
startup order
Using host networking
Starting up first test harness...
Starting up second test harness...
===== test session starts
=====
platform linux -- Python 3.10.12, pytest-8.4.2, pluggy-1.6.0 --
/app/.venv/bin/python
cachedir: .pytest_cache
metadata: {'Python': '3.10.12', 'Platform': 'Linux-6.6.87.2-microsoft-
standard-WSL2-x86_64-with-glibc2.35', 'Packages': {'pytest': '8.4.2',
'pluggy': '1.6.0'}, 'Plugins': {'json-report': '1.5.0', 'instafail': '0.5.0',
'metadata': '3.1.1'}}
rootdir: /app
configfile: pyproject.toml
plugins: json-report-1.5.0, instafail-0.5.0, metadata-3.1.1
collecting ...
----- live log collection -----
---
INFO      stomp.py:transport.py:786 established connection to host localhost,
port 61613
INFO      root:amq_asb_utils.py:89 Connected to ActiveMQ broker at
localhost:61613
INFO      root:amq_asb_utils.py:978 AMQ connection established on module
import
collected 1 item

compliance_test_harness/VI/test_activate_route.py::test_activate_route
----- live log setup -----
---
INFO      compliance_test_harness.conftest:conftest.py:54 Loaded
configuration from /app/test/configs/test-config.yaml successfully.
INFO      compliance_test_harness.conftest:conftest.py:55 Configuration
details: {
  "message_timeout_seconds": 6.0,
  "xml_generation_seed": 42,
  "disable_offboard_msg_wrapping": false,
  "offboard_consumer_config": {
    "wrapping_strategy": "tx"
  },
  "offboard_publisher_config": {
    "wrapping_strategy": "rx"
  },
  "platform_system_id": {
    "uuid_generation_namespace": {
      "namespace_uuid_name": "SUT",
      "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "1",
    "system_label": "1",

```

```

    "system_uuid_override": null,
    "system_uuid": "c5a0850c-282d-3211-b852-b92eb618a4ad"
  },
  "c2_system_id": {
    "uuid_generation_namespace": {
      "namespace_uuid_name": "SUT",
      "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "C2-1",
    "system_label": "C2 System",
    "system_uuid_override": null,
    "system_uuid": "a2f74db0-b0bb-33b4-b01b-744919f547bd"
  },
  "c2_system_id_2": {
    "uuid_generation_namespace": {
      "namespace_uuid_name": "SUT",
      "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "C2-2",
    "system_label": "C2 System 2",
    "system_uuid_override": null,
    "system_uuid": "5a64dc4d-d596-324c-88b8-7da2a0215651"
  },
  "peer_system_id": {
    "uuid_generation_namespace": {
      "namespace_uuid_name": "SUT",
      "namespace_uuid_id": "00000000-0000-0000-0000-000000000000"
    },
    "system_uuid_name": "2",
    "system_label": "Peer System",
    "system_uuid_override": null,
    "system_uuid": "31dbe31a-de46-39cc-92de-38d439ac2ab7"
  }
}
}
----- live log call -----
---
INFO      root:amq_asb_utils.py:196 Subscribed to /topic/FileLocation<None>
INFO      root:amq_asb_utils.py:196 Subscribed to /topic/FileMetadata<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_MissionPlanActivationCommandStatus<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_SystemNotification<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_MissionPlanActivationCommandStatus<None>
INFO      root:amq_asb_utils.py:196 Subscribed to
/topic/MA_MissionPlanActivationCommandStatus<None>
INFO
compliance_test_harness.VI.test_activate_route:test_activate_route.py:118
Sending MA_MissionPlanActivationCommand
INFO      common_utils.amq_asb_utils:amq_asb_utils.py:736 Publishing message
to VI without offboard wrapping
INFO      root:amq_asb_utils.py:770 Published message to
/topic/MA_MissionPlanActivationCommand<None>
INFO
compliance_test_harness.VI.test_activate_route:test_activate_route.py:149
Received MA_MissionPlanActivationCommandStatus

```

```

INFO
compliance_test_harness.VI.test_activate_route:test_activate_route.py:160
Received MA_MissionPlanActivationCommandStatus
INFO
compliance_test_harness.VI.test_activate_route:test_activate_route.py:177
Test completed successfully: Activate Route
PASSED
[100%]

===== 1 passed in 7.59s
=====
Found 2 passing test logs
PASSED (2/4)
===== LOGS =====
TEST: VI/test_activate_route.py
- VI_test_activate_route_20260417_151946.log (FAILED)
- MA_VI_test_activate_route_20260417_151946.log (FAILED)
- VI_test_activate_route_20260417_152028.log (PASSED)
- MA_VI_test_activate_route_20260417_152028.log (PASSED)

=====

/app/compliance_test_harness/VI/test_checksum_validation.py
/app/compliance_test_harness/MA/VI/test_checksum_validation.py
Running /app/compliance_test_harness/VI/test_checksum_validation.py
...

```

Inside of the test-harness-logs directory you should see two text files - one with results for the VI profile interactions and another with results for the MS profile interactions. Inside of test-harness-logs/old/ you will find .log container logs for individual test harness container runs. Since this test script does not know which order it needs to start the lifelines in to facilitate the test, it attempts both startup orders. As such, a total of 4 container log files are generated by this script for each sequence it executes - 2 for each direction.

If using the test-harness-runner.sh script, only the single .log file is generated for single test harness container that is started up. It will contain log output for all tests run by that container. These logs are rotated from the test-harness-logs/ to the test-harness-logs/old folder at the start of each run. The latest log when running this way should always be present in test-harness-logs/.

VI_<date>_<time>.txt

```

===== Test Results =====
PASSED (2/4) | TEST: VI/test_activate_route.py
PASSED (2/4) | TEST: VI/test_checksum_validation.py
PASSED (3/4) | TEST: VI/test_collision_avoidance.py
PASSED (2/4) | TEST: VI/test_control_by_curve_following.py
PASSED (2/4) | TEST: VI/test_control_by_hsa_csa_command.py
PASSED (2/4) | TEST: VI/test_control_by_waypoint_following.py
PASSED (2/4) | TEST: VI/test_control_mode_authorization.py
PASSED (2/4) | TEST: VI/test_convert_and_upload_route.py
PASSED (2/4) | TEST: VI/test_exchange_heartbeat_subsystem_status_reports.py
PASSED (4/4) | TEST: VI/test_intra_vehicle_comms_failure.py
PASSED (2/4) | TEST: VI/test_ma_failsafe.py

```

```

PASSED (2/4) | TEST: VI/test_ma_vi_command_task.py
PASSED (3/4) | TEST: VI/test_mechanical_damage_reporting.py
PASSED (3/4) | TEST: VI/test_modify_capabilities.py
PASSED (2/4) | TEST: VI/test_prepare_for_route_activation.py
PASSED (2/4) | TEST: VI/test_publish_control_status.py
PASSED (2/4) | TEST: VI/test_query_airfield_update.py
PASSED (2/4) | TEST: VI/test_query_for_missing_data.py
PASSED (2/4) | TEST: VI/test_query_route_plan.py
PASSED (2/4) | TEST: VI/test_receive_barometric_pressure.py
FAILED (0/4) | TEST: VI/test_receive_control_request.py
PASSED (2/4) | TEST: VI/test_receive_deactivate_route.py
PASSED (3/4) | TEST: VI/test_receive_execution_status.py
PASSED (3/4) | TEST: VI/test_receive_vehicle_performance_values.py
PASSED (3/4) | TEST: VI/test_receive_vehicle_state_data.py
PASSED (3/4) | TEST: VI/test_request_terrain_data.py
PASSED (4/4) | TEST: VI/test_route_plan_data_validation.py
PASSED (3/4) | TEST: VI/test_sensor_failure.py
PASSED (3/4) | TEST: VI/test_unpair_control_assignment.py
PASSED (2/4) | TEST: VI/test_update_c2_control_designations.py
PASSED (2/4) | TEST: VI/test_validate_release_envelope.py
PASSED (2/4) | TEST: VI/test_validate_route_plan.py
PASSED (3/4) | TEST: VI/test_vehicle_status_reporting.py
PASSED (3/4) | TEST: VI/test_vi_deactivate_route.py
PASSED (2/4) | TEST: VI/test_vi_responds_to_query_for_flight_capabilities.py
PASSED (3/4) | TEST: VI/test_vi_updates_to_cop.py
===== LOGS =====
TEST: VI/test_activate_route.py
- VI_test_activate_route_20260417_151946.log (FAILED)
- MA_VI_test_activate_route_20260417_151946.log (FAILED)
- VI_test_activate_route_20260417_152028.log (PASSED)
- MA_VI_test_activate_route_20260417_152028.log (PASSED)
TEST: VI/test_checksum_validation.py
- VI_test_checksum_validation_20260417_152059.log (FAILED)
- MA_VI_test_checksum_validation_20260417_152059.log (FAILED)
- VI_test_checksum_validation_20260417_152141.log (PASSED)
- MA_VI_test_checksum_validation_20260417_152141.log (PASSED)
TEST: VI/test_collision_avoidance.py
- VI_test_collision_avoidance_20260417_152213.log (PASSED)
- MA_VI_test_collision_avoidance_20260417_152213.log (PASSED)
- VI_test_collision_avoidance_20260417_152242.log (FAILED)
- MA_VI_test_collision_avoidance_20260417_152242.log (PASSED)
TEST: VI/test_control_by_curve_following.py
- VI_test_control_by_curve_following_20260417_152319.log (FAILED)
- MA_VI_test_control_by_curve_following_20260417_152319.log (FAILED)
- VI_test_control_by_curve_following_20260417_152401.log (PASSED)
- MA_VI_test_control_by_curve_following_20260417_152401.log (PASSED)
TEST: VI/test_control_by_hsa_csa_command.py
- VI_test_control_by_hsa_csa_command_20260417_152433.log (FAILED)
- MA_VI_test_control_by_hsa_csa_command_20260417_152433.log (FAILED)
- VI_test_control_by_hsa_csa_command_20260417_152515.log (PASSED)
- MA_VI_test_control_by_hsa_csa_command_20260417_152515.log (PASSED)
TEST: VI/test_control_by_waypoint_following.py
- VI_test_control_by_waypoint_following_20260417_152547.log (FAILED)
- MA_VI_test_control_by_waypoint_following_20260417_152547.log (FAILED)
- VI_test_control_by_waypoint_following_20260417_152629.log (PASSED)
- MA_VI_test_control_by_waypoint_following_20260417_152629.log (PASSED)
TEST: VI/test_control_mode_authorization.py

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- VI_test_control_mode_authorization_20260417_152700.log (PASSED)
- MA_VI_test_control_mode_authorization_20260417_152700.log (PASSED)
- VI_test_control_mode_authorization_20260417_152729.log (FAILED)
- MA_VI_test_control_mode_authorization_20260417_152729.log (FAILED)

TEST: VI/test_convert_and_upload_route.py

- VI_test_convert_and_upload_route_20260417_152814.log (FAILED)
- MA_VI_test_convert_and_upload_route_20260417_152814.log (FAILED)
- VI_test_convert_and_upload_route_20260417_152856.log (PASSED)
- MA_VI_test_convert_and_upload_route_20260417_152856.log (PASSED)

TEST: VI/test_exchange_heartbeat_subsystem_status_reports.py

- VI_test_exchange_heartbeat_subsystem_status_reports_20260417_152927.log (FAILED)

-

MA_VI_test_exchange_heartbeat_subsystem_status_reports_20260417_152927.log (FAILED)

- VI_test_exchange_heartbeat_subsystem_status_reports_20260417_153010.log (PASSED)

-

MA_VI_test_exchange_heartbeat_subsystem_status_reports_20260417_153010.log (PASSED)

TEST: VI/test_intra_vehicle_comms_failure.py

- VI_test_intra_vehicle_comms_failure_20260417_153041.log (PASSED)
- MA_VI_test_intra_vehicle_comms_failure_20260417_153041.log (PASSED)
- VI_test_intra_vehicle_comms_failure_20260417_153109.log (PASSED)
- MA_VI_test_intra_vehicle_comms_failure_20260417_153109.log (PASSED)

TEST: VI/test_ma_failsafe.py

- VI_test_ma_failsafe_20260417_153140.log (FAILED)
- MA_VI_test_ma_failsafe_20260417_153140.log (FAILED)
- VI_test_ma_failsafe_20260417_153222.log (PASSED)
- MA_VI_test_ma_failsafe_20260417_153222.log (PASSED)

TEST: VI/test_ma_vi_command_task.py

- VI_test_ma_vi_command_task_20260417_153253.log (FAILED)
- MA_VI_test_ma_vi_command_task_20260417_153253.log (FAILED)
- VI_test_ma_vi_command_task_20260417_153337.log (PASSED)
- MA_VI_test_ma_vi_command_task_20260417_153337.log (PASSED)

TEST: VI/test_mechanical_damage_reporting.py

- VI_test_mechanical_damage_reporting_20260417_153409.log (PASSED)
- MA_VI_test_mechanical_damage_reporting_20260417_153409.log (PASSED)
- VI_test_mechanical_damage_reporting_20260417_153439.log (FAILED)
- MA_VI_test_mechanical_damage_reporting_20260417_153439.log (PASSED)

TEST: VI/test_modify_capabilities.py

- VI_test_modify_capabilities_20260417_153515.log (PASSED)
- MA_VI_test_modify_capabilities_20260417_153515.log (PASSED)
- VI_test_modify_capabilities_20260417_153545.log (FAILED)
- MA_VI_test_modify_capabilities_20260417_153545.log (PASSED)

TEST: VI/test_prepare_for_route_activation.py

- VI_test_prepare_for_route_activation_20260417_153622.log (FAILED)
- MA_VI_test_prepare_for_route_activation_20260417_153622.log (FAILED)
- VI_test_prepare_for_route_activation_20260417_153704.log (PASSED)
- MA_VI_test_prepare_for_route_activation_20260417_153704.log (PASSED)

TEST: VI/test_publish_control_status.py

- VI_test_publish_control_status_20260417_153736.log (PASSED)
- MA_VI_test_publish_control_status_20260417_153736.log (PASSED)
- VI_test_publish_control_status_20260417_153807.log (FAILED)
- MA_VI_test_publish_control_status_20260417_153807.log (FAILED)

TEST: VI/test_query_airfield_update.py

- VI_test_query_airfield_update_20260417_153852.log (FAILED)

- MA_VI_test_query_airfield_update_20260417_153852.log (FAILED)
- VI_test_query_airfield_update_20260417_153934.log (PASSED)
- MA_VI_test_query_airfield_update_20260417_153934.log (PASSED)

TEST: VI/test_query_for_missing_data.py

- VI_test_query_for_missing_data_20260417_154006.log (FAILED)
- MA_VI_test_query_for_missing_data_20260417_154006.log (FAILED)
- VI_test_query_for_missing_data_20260417_154048.log (PASSED)
- MA_VI_test_query_for_missing_data_20260417_154048.log (PASSED)

TEST: VI/test_query_route_plan.py

- VI_test_query_route_plan_20260417_154120.log (FAILED)
- MA_VI_test_query_route_plan_20260417_154120.log (FAILED)
- VI_test_query_route_plan_20260417_154203.log (PASSED)
- MA_VI_test_query_route_plan_20260417_154203.log (PASSED)

TEST: VI/test_receive_barometric_pressure.py

- VI_test_receive_barometric_pressure_20260417_154234.log (FAILED)
- MA_VI_test_receive_barometric_pressure_20260417_154234.log (FAILED)
- VI_test_receive_barometric_pressure_20260417_154317.log (PASSED)
- MA_VI_test_receive_barometric_pressure_20260417_154317.log (PASSED)

TEST: VI/test_receive_control_request.py

- VI_test_receive_control_request_20260417_154348.log (FAILED)
- MA_VI_test_receive_control_request_20260417_154348.log (FAILED)
- VI_test_receive_control_request_20260417_154432.log (FAILED)
- MA_VI_test_receive_control_request_20260417_154432.log (FAILED)

TEST: VI/test_receive_deactivate_route.py

- VI_test_receive_deactivate_route_20260417_154517.log (FAILED)
- MA_VI_test_receive_deactivate_route_20260417_154517.log (FAILED)
- VI_test_receive_deactivate_route_20260417_154600.log (PASSED)
- MA_VI_test_receive_deactivate_route_20260417_154600.log (PASSED)

TEST: VI/test_receive_execution_status.py

- VI_test_receive_execution_status_20260417_154633.log (PASSED)
- MA_VI_test_receive_execution_status_20260417_154633.log (PASSED)
- VI_test_receive_execution_status_20260417_154706.log (FAILED)
- MA_VI_test_receive_execution_status_20260417_154706.log (PASSED)

TEST: VI/test_receive_vehicle_performance_values.py

- VI_test_receive_vehicle_performance_values_20260417_154743.log (PASSED)
- MA_VI_test_receive_vehicle_performance_values_20260417_154743.log (PASSED)
- VI_test_receive_vehicle_performance_values_20260417_154812.log (FAILED)
- MA_VI_test_receive_vehicle_performance_values_20260417_154812.log (PASSED)

TEST: VI/test_receive_vehicle_state_data.py

- VI_test_receive_vehicle_state_data_20260417_154848.log (PASSED)
- MA_VI_test_receive_vehicle_state_data_20260417_154848.log (PASSED)
- VI_test_receive_vehicle_state_data_20260417_154917.log (FAILED)
- MA_VI_test_receive_vehicle_state_data_20260417_154917.log (PASSED)

TEST: VI/test_request_terrain_data.py

- VI_test_request_terrain_data_20260417_154954.log (PASSED)
- MA_VI_test_request_terrain_data_20260417_154954.log (FAILED)
- VI_test_request_terrain_data_20260417_155027.log (PASSED)
- MA_VI_test_request_terrain_data_20260417_155027.log (PASSED)

TEST: VI/test_route_plan_data_validation.py

- VI_test_route_plan_data_validation_20260417_155059.log (PASSED)
- MA_VI_test_route_plan_data_validation_20260417_155059.log (PASSED)
- VI_test_route_plan_data_validation_20260417_155127.log (PASSED)
- MA_VI_test_route_plan_data_validation_20260417_155127.log (PASSED)

TEST: VI/test_sensor_failure.py

- VI_test_sensor_failure_20260417_155157.log (PASSED)

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- MA_VI_test_sensor_failure_20260417_155157.log (PASSED)
- VI_test_sensor_failure_20260417_155227.log (FAILED)
- MA_VI_test_sensor_failure_20260417_155227.log (PASSED)
TEST: VI/test_unpair_control_assignment.py
- VI_test_unpair_control_assignment_20260417_155302.log (PASSED)
- MA_VI_test_unpair_control_assignment_20260417_155302.log (PASSED)
- VI_test_unpair_control_assignment_20260417_155332.log (FAILED)
- MA_VI_test_unpair_control_assignment_20260417_155332.log (PASSED)
TEST: VI/test_update_c2_control_designations.py
- VI_test_update_c2_control_designations_20260417_155408.log (FAILED)
- MA_VI_test_update_c2_control_designations_20260417_155408.log (FAILED)
- VI_test_update_c2_control_designations_20260417_155450.log (PASSED)
- MA_VI_test_update_c2_control_designations_20260417_155450.log (PASSED)
TEST: VI/test_validate_release_envelope.py
- VI_test_validate_release_envelope_20260417_155521.log (FAILED)
- MA_VI_test_validate_release_envelope_20260417_155521.log (FAILED)
- VI_test_validate_release_envelope_20260417_155604.log (PASSED)
- MA_VI_test_validate_release_envelope_20260417_155604.log (PASSED)
TEST: VI/test_validate_route_plan.py
- VI_test_validate_route_plan_20260417_155637.log (FAILED)
- MA_VI_test_validate_route_plan_20260417_155637.log (FAILED)
- VI_test_validate_route_plan_20260417_155720.log (PASSED)
- MA_VI_test_validate_route_plan_20260417_155720.log (PASSED)
TEST: VI/test_vehicle_status_reporting.py
- VI_test_vehicle_status_reporting_20260417_155752.log (PASSED)
- MA_VI_test_vehicle_status_reporting_20260417_155752.log (PASSED)
- VI_test_vehicle_status_reporting_20260417_155823.log (FAILED)
- MA_VI_test_vehicle_status_reporting_20260417_155823.log (PASSED)
TEST: VI/test_vi_deactivate_route.py
- VI_test_vi_deactivate_route_20260417_155900.log (PASSED)
- MA_VI_test_vi_deactivate_route_20260417_155900.log (PASSED)
- VI_test_vi_deactivate_route_20260417_155929.log (FAILED)
- MA_VI_test_vi_deactivate_route_20260417_155929.log (PASSED)
TEST: VI/test_vi_responds_to_query_for_flight_capabilities.py
- VI_test_vi_responds_to_query_for_flight_capabilities_20260417_160006.log
(FAILED)
-
MA_VI_test_vi_responds_to_query_for_flight_capabilities_20260417_160006.log
(FAILED)
- VI_test_vi_responds_to_query_for_flight_capabilities_20260417_160053.log
(PASSED)
-
MA_VI_test_vi_responds_to_query_for_flight_capabilities_20260417_160053.log
(PASSED)
TEST: VI/test_vi_updates_to_cop.py
- VI_test_vi_updates_to_cop_20260417_160134.log (PASSED)
- MA_VI_test_vi_updates_to_cop_20260417_160134.log (PASSED)
- VI_test_vi_updates_to_cop_20260417_160205.log (FAILED)
- MA_VI_test_vi_updates_to_cop_20260417_160205.log (PASSED)
===== PASSED:35 | FAILED:1 =====

```

MS_<date>_<time>.txt

===== Test Results =====

```

PASSED (2/4) | TEST: MS/test_abort_release_sequence.py
PASSED (3/4) | TEST: MS/test_advertise_sensor_amti_capability.py
PASSED (3/4) | TEST: MS/test_advertise_sensor_esm_capability.py
PASSED (3/4) | TEST: MS/test_advertise_sensor_po_capability.py

```

```

PASSED (2/4) | TEST: MS/test_command_sensor_amti_capability.py
PASSED (2/4) | TEST: MS/test_command_sensor_esm_capability.py
PASSED (2/4) | TEST: MS/test_command_sensor_po_capability.py
PASSED (4/4) | TEST: MS/test_command_squawk.py
PASSED (3/4) | TEST: MS/test_communications_lost.py
PASSED (2/4) | TEST: MS/test_configure_communications_terminal.py
PASSED (3/4) | TEST: MS/test_confirm_threat_id.py
PASSED (3/4) | TEST:
MS/test_distribute_single_sensor_track_data_to_perform_local_fusion.py
PASSED (3/4) | TEST: MS/test_establish_communications_terminal.py
PASSED (2/4) | TEST: MS/test_exchange_dlz.py
PASSED (2/4) | TEST: MS/test_exchange_wez.py
PASSED (2/4) | TEST: MS/test_fire_command_exchange.py
PASSED (4/4) | TEST: MS/test_intra_vehicle_comms_failure_ms.py
PASSED (2/4) | TEST: MS/test_jettison_release_sequence.py
PASSED (3/4) | TEST: MS/test_logging_failure.py
PASSED (3/4) | TEST: MS/test_maintain_communication_terminal.py
PASSED (2/4) | TEST: MS/test_mission_store_release_initialization_sequence.py
PASSED (3/4) | TEST: MS/test_mission_store_status_exchange.py
PASSED (3/4) | TEST: MS/test_mission_systems_capability_heartbeat.py
PASSED (3/4) | TEST: MS/test_ms_comms_status_updates_to_cop.py
PASSED (2/4) | TEST: MS/test_ms_responds_to_query_for_capabilities.py
PASSED (3/4) | TEST: MS/test_ms_sensor_amti_status_updates_to_cop.py
PASSED (4/4) | TEST: MS/test_ms_sensor_po_status_updates_to_cop.py
PASSED (3/4) | TEST: MS/test_ms_weapon_status_updates_to_cop.py
PASSED (3/4) | TEST: MS/test_payload_management_failure.py
PASSED (3/4) | TEST: MS/test_publish_comm_team_report.py
FAILED (0/4) | TEST: MS/test_publish_external_weather_reports.py
PASSED (3/4) | TEST: MS/test_publish_network_endpoint_availability.py
PASSED (3/4) | TEST:
MS/test_publish_periodic_passive_optical_sensor_configuration.py
PASSED (2/4) | TEST:
MS/test_receive_changes_to_stores_primary_mission_plan_data.py
PASSED (3/4) | TEST: MS/test_receive_communication_error_report.py
PASSED (3/4) | TEST: MS/test_receive_encoded_payload_message.py
PASSED (3/4) | TEST: MS/test_receive_message_subscription_status.py
PASSED (3/4) | TEST:
MS/test_receive_periodic_health_status_bit_information.py
FAILED (0/4) | TEST: MS/test_receive_plans.py
PASSED (3/4) | TEST: MS/test_receive_processed_track.py
PASSED (2/4) | TEST: MS/test_release_consent_sequence.py
PASSED (3/4) | TEST: MS/test_report_sensor_amti_capability_state.py
PASSED (3/4) | TEST: MS/test_report_sensor_esm_capability_state.py
PASSED (3/4) | TEST: MS/test_report_sensor_po_capability_state.py
PASSED (3/4) | TEST: MS/test_report_weather_radar_capability_observations.py
PASSED (2/4) | TEST: MS/test_request_communication_asset_status.py
PASSED (2/4) | TEST:
MS/test_request_mission_system_health_status_bit_information.py
PASSED (4/4) | TEST: MS/test_request_pnt_navigation_data.py
PASSED (2/4) | TEST: MS/test_send_encoded_payload_message.py
PASSED (3/4) | TEST: MS/test_sensor_failure.py
PASSED (3/4) | TEST: MS/test_store_inventory_and_descriptions_exchange.py
PASSED (2/4) | TEST: MS/test_subscribe_to_message_topic.py
PASSED (2/4) | TEST:
MS/test_update_communication_manager_communication_plan.py
===== LOGS =====
TEST: MS/test_abort_release_sequence.py

```

- MS_test_abort_release_sequence_20260417_160242.log (FAILED)
- MA_MS_test_abort_release_sequence_20260417_160242.log (FAILED)
- MS_test_abort_release_sequence_20260417_160324.log (PASSED)
- MA_MS_test_abort_release_sequence_20260417_160324.log (PASSED)

TEST: MS/test_advertise_sensor_amti_capability.py

- MS_test_advertise_sensor_amti_capability_20260417_160355.log (PASSED)
- MA_MS_test_advertise_sensor_amti_capability_20260417_160355.log (PASSED)
- MS_test_advertise_sensor_amti_capability_20260417_160425.log (FAILED)
- MA_MS_test_advertise_sensor_amti_capability_20260417_160425.log (PASSED)

TEST: MS/test_advertise_sensor_esm_capability.py

- MS_test_advertise_sensor_esm_capability_20260417_160501.log (PASSED)
- MA_MS_test_advertise_sensor_esm_capability_20260417_160501.log (PASSED)
- MS_test_advertise_sensor_esm_capability_20260417_160531.log (FAILED)
- MA_MS_test_advertise_sensor_esm_capability_20260417_160531.log (PASSED)

TEST: MS/test_advertise_sensor_po_capability.py

- MS_test_advertise_sensor_po_capability_20260417_160607.log (PASSED)
- MA_MS_test_advertise_sensor_po_capability_20260417_160607.log (PASSED)
- MS_test_advertise_sensor_po_capability_20260417_160636.log (FAILED)
- MA_MS_test_advertise_sensor_po_capability_20260417_160636.log (PASSED)

TEST: MS/test_command_sensor_amti_capability.py

- MS_test_command_sensor_amti_capability_20260417_160711.log (FAILED)
- MA_MS_test_command_sensor_amti_capability_20260417_160711.log (FAILED)
- MS_test_command_sensor_amti_capability_20260417_160752.log (PASSED)
- MA_MS_test_command_sensor_amti_capability_20260417_160752.log (PASSED)

TEST: MS/test_command_sensor_esm_capability.py

- MS_test_command_sensor_esm_capability_20260417_160823.log (FAILED)
- MA_MS_test_command_sensor_esm_capability_20260417_160823.log (FAILED)
- MS_test_command_sensor_esm_capability_20260417_160905.log (PASSED)
- MA_MS_test_command_sensor_esm_capability_20260417_160905.log (PASSED)

TEST: MS/test_command_sensor_po_capability.py

- MS_test_command_sensor_po_capability_20260417_160935.log (FAILED)
- MA_MS_test_command_sensor_po_capability_20260417_160935.log (FAILED)
- MS_test_command_sensor_po_capability_20260417_161017.log (PASSED)
- MA_MS_test_command_sensor_po_capability_20260417_161017.log (PASSED)

TEST: MS/test_command_squawk.py

- MS_test_command_squawk_20260417_161047.log (PASSED)
- MA_MS_test_command_squawk_20260417_161047.log (PASSED)
- MS_test_command_squawk_20260417_161114.log (PASSED)
- MA_MS_test_command_squawk_20260417_161114.log (PASSED)

TEST: MS/test_communications_lost.py

- MS_test_communications_lost_20260417_161143.log (PASSED)
- MA_MS_test_communications_lost_20260417_161143.log (PASSED)
- MS_test_communications_lost_20260417_161212.log (FAILED)
- MA_MS_test_communications_lost_20260417_161212.log (PASSED)

TEST: MS/test_configure_communications_terminal.py

- MS_test_configure_communications_terminal_20260417_161249.log (FAILED)
- MA_MS_test_configure_communications_terminal_20260417_161249.log (FAILED)
- MS_test_configure_communications_terminal_20260417_161331.log (PASSED)
- MA_MS_test_configure_communications_terminal_20260417_161331.log (PASSED)

TEST: MS/test_confirm_threat_id.py

- MS_test_confirm_threat_id_20260417_161403.log (PASSED)
- MA_MS_test_confirm_threat_id_20260417_161403.log (PASSED)
- MS_test_confirm_threat_id_20260417_161434.log (FAILED)
- MA_MS_test_confirm_threat_id_20260417_161434.log (PASSED)

TEST: MS/test_distribute_single_sensor_track_data_to_perform_local_fusion.py

```

-
MS_test_distribute_single_sensor_track_data_to_perform_local_fusion_20260417_
161510.log (PASSED)
-
MA_MS_test_distribute_single_sensor_track_data_to_perform_local_fusion_202604
17_161510.log (PASSED)
-
MS_test_distribute_single_sensor_track_data_to_perform_local_fusion_20260417_
161539.log (FAILED)
-
MA_MS_test_distribute_single_sensor_track_data_to_perform_local_fusion_202604
17_161539.log (PASSED)
TEST: MS/test_establish_communications_terminal.py
- MS_test_establish_communications_terminal_20260417_161615.log (PASSED)
- MA_MS_test_establish_communications_terminal_20260417_161615.log (PASSED)
- MS_test_establish_communications_terminal_20260417_161645.log (FAILED)
- MA_MS_test_establish_communications_terminal_20260417_161645.log (PASSED)
TEST: MS/test_exchange_dlz.py
- MS_test_exchange_dlz_20260417_161721.log (FAILED)
- MA_MS_test_exchange_dlz_20260417_161721.log (FAILED)
- MS_test_exchange_dlz_20260417_161804.log (PASSED)
- MA_MS_test_exchange_dlz_20260417_161804.log (PASSED)
TEST: MS/test_exchange_wez.py
- MS_test_exchange_wez_20260417_161837.log (FAILED)
- MA_MS_test_exchange_wez_20260417_161837.log (FAILED)
- MS_test_exchange_wez_20260417_161921.log (PASSED)
- MA_MS_test_exchange_wez_20260417_161921.log (PASSED)
TEST: MS/test_fire_command_exchange.py
- MS_test_fire_command_exchange_20260417_161953.log (FAILED)
- MA_MS_test_fire_command_exchange_20260417_161953.log (FAILED)
- MS_test_fire_command_exchange_20260417_162036.log (PASSED)
- MA_MS_test_fire_command_exchange_20260417_162036.log (PASSED)
TEST: MS/test_intra_vehicle_comms_failure_ms.py
- MS_test_intra_vehicle_comms_failure_ms_20260417_162107.log (PASSED)
- MA_MS_test_intra_vehicle_comms_failure_ms_20260417_162107.log (PASSED)
- MS_test_intra_vehicle_comms_failure_ms_20260417_162135.log (PASSED)
- MA_MS_test_intra_vehicle_comms_failure_ms_20260417_162135.log (PASSED)
TEST: MS/test_jettison_release_sequence.py
- MS_test_jettison_release_sequence_20260417_162206.log (FAILED)
- MA_MS_test_jettison_release_sequence_20260417_162206.log (FAILED)
- MS_test_jettison_release_sequence_20260417_162248.log (PASSED)
- MA_MS_test_jettison_release_sequence_20260417_162248.log (PASSED)
TEST: MS/test_logging_failure.py
- MS_test_logging_failure_20260417_162320.log (PASSED)
- MA_MS_test_logging_failure_20260417_162320.log (PASSED)
- MS_test_logging_failure_20260417_162350.log (FAILED)
- MA_MS_test_logging_failure_20260417_162350.log (PASSED)
TEST: MS/test_maintain_communication_terminal.py
- MS_test_maintain_communication_terminal_20260417_162426.log (PASSED)
- MA_MS_test_maintain_communication_terminal_20260417_162426.log (PASSED)
- MS_test_maintain_communication_terminal_20260417_162456.log (FAILED)
- MA_MS_test_maintain_communication_terminal_20260417_162456.log (PASSED)
TEST: MS/test_mission_store_release_initialization_sequence.py
- MS_test_mission_store_release_initialization_sequence_20260417_162532.log
(PASSED)

```

```

-
MA_MS_test_mission_store_release_initialization_sequence_20260417_162532.log
(PASSED)
- MS_test_mission_store_release_initialization_sequence_20260417_162601.log
(FAILED)
-
MA_MS_test_mission_store_release_initialization_sequence_20260417_162601.log
(FAILED)
TEST: MS/test_mission_store_status_exchange.py
- MS_test_mission_store_status_exchange_20260417_162647.log (PASSED)
- MA_MS_test_mission_store_status_exchange_20260417_162647.log (PASSED)
- MS_test_mission_store_status_exchange_20260417_162716.log (FAILED)
- MA_MS_test_mission_store_status_exchange_20260417_162716.log (PASSED)
TEST: MS/test_mission_systems_capability_heartbeat.py
- MS_test_mission_systems_capability_heartbeat_20260417_162752.log (PASSED)
- MA_MS_test_mission_systems_capability_heartbeat_20260417_162752.log
(PASSED)
- MS_test_mission_systems_capability_heartbeat_20260417_162822.log (FAILED)
- MA_MS_test_mission_systems_capability_heartbeat_20260417_162822.log
(PASSED)
TEST: MS/test_ms_comms_status_updates_to_cop.py
- MS_test_ms_comms_status_updates_to_cop_20260417_162857.log (PASSED)
- MA_MS_test_ms_comms_status_updates_to_cop_20260417_162857.log (PASSED)
- MS_test_ms_comms_status_updates_to_cop_20260417_162926.log (FAILED)
- MA_MS_test_ms_comms_status_updates_to_cop_20260417_162926.log (PASSED)
TEST: MS/test_ms_responds_to_query_for_capabilities.py
- MS_test_ms_responds_to_query_for_capabilities_20260417_163001.log
(FAILED)
- MA_MS_test_ms_responds_to_query_for_capabilities_20260417_163001.log
(FAILED)
- MS_test_ms_responds_to_query_for_capabilities_20260417_163042.log
(PASSED)
- MA_MS_test_ms_responds_to_query_for_capabilities_20260417_163042.log
(PASSED)
TEST: MS/test_ms_sensor_amti_status_updates_to_cop.py
- MS_test_ms_sensor_amti_status_updates_to_cop_20260417_163113.log (PASSED)
- MA_MS_test_ms_sensor_amti_status_updates_to_cop_20260417_163113.log
(PASSED)
- MS_test_ms_sensor_amti_status_updates_to_cop_20260417_163141.log (FAILED)
- MA_MS_test_ms_sensor_amti_status_updates_to_cop_20260417_163141.log
(PASSED)
TEST: MS/test_ms_sensor_po_status_updates_to_cop.py
- MS_test_ms_sensor_po_status_updates_to_cop_20260417_163217.log (PASSED)
- MA_MS_test_ms_sensor_po_status_updates_to_cop_20260417_163217.log
(PASSED)
- MS_test_ms_sensor_po_status_updates_to_cop_20260417_163244.log (PASSED)
- MA_MS_test_ms_sensor_po_status_updates_to_cop_20260417_163244.log
(PASSED)
TEST: MS/test_ms_weapon_status_updates_to_cop.py
- MS_test_ms_weapon_status_updates_to_cop_20260417_163313.log (PASSED)
- MA_MS_test_ms_weapon_status_updates_to_cop_20260417_163313.log (PASSED)
- MS_test_ms_weapon_status_updates_to_cop_20260417_163343.log (FAILED)
- MA_MS_test_ms_weapon_status_updates_to_cop_20260417_163343.log (PASSED)
TEST: MS/test_payload_management_failure.py
- MS_test_payload_management_failure_20260417_163420.log (PASSED)
- MA_MS_test_payload_management_failure_20260417_163420.log (PASSED)
- MS_test_payload_management_failure_20260417_163450.log (FAILED)

```

```

- MA_MS_test_payload_management_failure_20260417_163450.log (PASSED)
TEST: MS/test_publish_comm_team_report.py
- MS_test_publish_comm_team_report_20260417_163526.log (PASSED)
- MA_MS_test_publish_comm_team_report_20260417_163526.log (PASSED)
- MS_test_publish_comm_team_report_20260417_163556.log (FAILED)
- MA_MS_test_publish_comm_team_report_20260417_163556.log (PASSED)
TEST: MS/test_publish_external_weather_reports.py
- MS_test_publish_external_weather_reports_20260417_163633.log (FAILED)
- MA_MS_test_publish_external_weather_reports_20260417_163633.log (FAILED)
- MS_test_publish_external_weather_reports_20260417_163707.log (FAILED)
- MA_MS_test_publish_external_weather_reports_20260417_163707.log (FAILED)
TEST: MS/test_publish_network_endpoint_availability.py
- MS_test_publish_network_endpoint_availability_20260417_163748.log
(PASSED)
- MA_MS_test_publish_network_endpoint_availability_20260417_163748.log
(PASSED)
- MS_test_publish_network_endpoint_availability_20260417_163817.log
(FAILED)
- MA_MS_test_publish_network_endpoint_availability_20260417_163817.log
(PASSED)
TEST: MS/test_publish_periodic_passive_optical_sensor_configuration.py
-
MS_test_publish_periodic_passive_optical_sensor_configuration_20260417_163852
.log (PASSED)
-
MA_MS_test_publish_periodic_passive_optical_sensor_configuration_20260417_163
852.log (PASSED)
-
MS_test_publish_periodic_passive_optical_sensor_configuration_20260417_163921
.log (FAILED)
-
MA_MS_test_publish_periodic_passive_optical_sensor_configuration_20260417_163
921.log (PASSED)
TEST: MS/test_receive_changes_to_stores_primary_mission_plan_data.py
-
MS_test_receive_changes_to_stores_primary_mission_plan_data_20260417_163956.1
og (PASSED)
-
MA_MS_test_receive_changes_to_stores_primary_mission_plan_data_20260417_16395
6.log (PASSED)
-
MS_test_receive_changes_to_stores_primary_mission_plan_data_20260417_164025.1
og (FAILED)
-
MA_MS_test_receive_changes_to_stores_primary_mission_plan_data_20260417_16402
5.log (FAILED)
TEST: MS/test_receive_communication_error_report.py
- MS_test_receive_communication_error_report_20260417_164109.log (PASSED)
- MA_MS_test_receive_communication_error_report_20260417_164109.log
(PASSED)
- MS_test_receive_communication_error_report_20260417_164138.log (FAILED)
- MA_MS_test_receive_communication_error_report_20260417_164138.log
(PASSED)
TEST: MS/test_receive_encoded_payload_message.py
- MS_test_receive_encoded_payload_message_20260417_164213.log (PASSED)
- MA_MS_test_receive_encoded_payload_message_20260417_164213.log (PASSED)
- MS_test_receive_encoded_payload_message_20260417_164241.log (FAILED)

```

```

- MA_MS_test_receive_encoded_payload_message_20260417_164241.log (PASSED)
TEST: MS/test_receive_message_subscription_status.py
- MS_test_receive_message_subscription_status_20260417_164318.log (PASSED)
- MA_MS_test_receive_message_subscription_status_20260417_164318.log
(PASSED)
- MS_test_receive_message_subscription_status_20260417_164348.log (FAILED)
- MA_MS_test_receive_message_subscription_status_20260417_164348.log
(PASSED)
TEST: MS/test_receive_periodic_health_status_bit_information.py
-
MS_test_receive_periodic_health_status_bit_information_20260417_164424.log
(PASSED)
-
MA_MS_test_receive_periodic_health_status_bit_information_20260417_164424.log
(PASSED)
-
MS_test_receive_periodic_health_status_bit_information_20260417_164454.log
(FAILED)
-
MA_MS_test_receive_periodic_health_status_bit_information_20260417_164454.log
(PASSED)
TEST: MS/test_receive_plans.py
- MS_test_receive_plans_20260417_164531.log (FAILED)
- MA_MS_test_receive_plans_20260417_164531.log (FAILED)
- MS_test_receive_plans_20260417_164605.log (FAILED)
- MA_MS_test_receive_plans_20260417_164605.log (FAILED)
TEST: MS/test_receive_processed_track.py
- MS_test_receive_processed_track_20260417_164646.log (PASSED)
- MA_MS_test_receive_processed_track_20260417_164646.log (PASSED)
- MS_test_receive_processed_track_20260417_164716.log (FAILED)
- MA_MS_test_receive_processed_track_20260417_164716.log (PASSED)
TEST: MS/test_release_consent_sequence.py
- MS_test_release_consent_sequence_20260417_164752.log (FAILED)
- MA_MS_test_release_consent_sequence_20260417_164752.log (FAILED)
- MS_test_release_consent_sequence_20260417_164834.log (PASSED)
- MA_MS_test_release_consent_sequence_20260417_164834.log (PASSED)
TEST: MS/test_report_sensor_amti_capability_state.py
- MS_test_report_sensor_amti_capability_state_20260417_164907.log (PASSED)
- MA_MS_test_report_sensor_amti_capability_state_20260417_164907.log
(PASSED)
- MS_test_report_sensor_amti_capability_state_20260417_164936.log (FAILED)
- MA_MS_test_report_sensor_amti_capability_state_20260417_164936.log
(PASSED)
TEST: MS/test_report_sensor_esm_capability_state.py
- MS_test_report_sensor_esm_capability_state_20260417_165012.log (PASSED)
- MA_MS_test_report_sensor_esm_capability_state_20260417_165012.log
(PASSED)
- MS_test_report_sensor_esm_capability_state_20260417_165042.log (FAILED)
- MA_MS_test_report_sensor_esm_capability_state_20260417_165042.log
(PASSED)
TEST: MS/test_report_sensor_po_capability_state.py
- MS_test_report_sensor_po_capability_state_20260417_165118.log (PASSED)
- MA_MS_test_report_sensor_po_capability_state_20260417_165118.log (PASSED)
- MS_test_report_sensor_po_capability_state_20260417_165147.log (FAILED)
- MA_MS_test_report_sensor_po_capability_state_20260417_165147.log (PASSED)
TEST: MS/test_report_weather_radar_capability_observations.py

```



```

- MS_test_report_weather_radar_capability_observations_20260417_165224.log
(PASSED)
-
MA_MS_test_report_weather_radar_capability_observations_20260417_165224.log
(PASSED)
- MS_test_report_weather_radar_capability_observations_20260417_165253.log
(FAILED)
-
MA_MS_test_report_weather_radar_capability_observations_20260417_165253.log
(PASSED)
TEST: MS/test_request_communication_asset_status.py
- MS_test_request_communication_asset_status_20260417_165329.log (FAILED)
- MA_MS_test_request_communication_asset_status_20260417_165329.log
(FAILED)
- MS_test_request_communication_asset_status_20260417_165411.log (PASSED)
- MA_MS_test_request_communication_asset_status_20260417_165411.log
(PASSED)
TEST: MS/test_request_mission_system_health_status_bit_information.py
-
MS_test_request_mission_system_health_status_bit_information_20260417_165442.
log (FAILED)
-
MA_MS_test_request_mission_system_health_status_bit_information_20260417_1654
42.log (FAILED)
-
MS_test_request_mission_system_health_status_bit_information_20260417_165524.
log (PASSED)
-
MA_MS_test_request_mission_system_health_status_bit_information_20260417_1655
24.log (PASSED)
TEST: MS/test_request_pnt_navigation_data.py
- MS_test_request_pnt_navigation_data_20260417_165555.log (PASSED)
- MA_MS_test_request_pnt_navigation_data_20260417_165555.log (PASSED)
- MS_test_request_pnt_navigation_data_20260417_165622.log (PASSED)
- MA_MS_test_request_pnt_navigation_data_20260417_165622.log (PASSED)
TEST: MS/test_send_encoded_payload_message.py
- MS_test_send_encoded_payload_message_20260417_165652.log (FAILED)
- MA_MS_test_send_encoded_payload_message_20260417_165652.log (FAILED)
- MS_test_send_encoded_payload_message_20260417_165734.log (PASSED)
- MA_MS_test_send_encoded_payload_message_20260417_165734.log (PASSED)
TEST: MS/test_sensor_failure.py
- MS_test_sensor_failure_20260417_165805.log (PASSED)
- MA_MS_test_sensor_failure_20260417_165805.log (PASSED)
- MS_test_sensor_failure_20260417_165834.log (FAILED)
- MA_MS_test_sensor_failure_20260417_165834.log (PASSED)
TEST: MS/test_store_inventory_and_descriptions_exchange.py
- MS_test_store_inventory_and_descriptions_exchange_20260417_165911.log
(PASSED)
- MA_MS_test_store_inventory_and_descriptions_exchange_20260417_165911.log
(PASSED)
- MS_test_store_inventory_and_descriptions_exchange_20260417_165940.log
(FAILED)
- MA_MS_test_store_inventory_and_descriptions_exchange_20260417_165940.log
(PASSED)
TEST: MS/test_subscribe_to_message_topic.py
- MS_test_subscribe_to_message_topic_20260417_170016.log (FAILED)
- MA_MS_test_subscribe_to_message_topic_20260417_170016.log (FAILED)

```

```

- MS_test_subscribe_to_message_topic_20260417_170058.log (PASSED)
- MA_MS_test_subscribe_to_message_topic_20260417_170058.log (PASSED)
TEST: MS/test_update_communication_manager_communication_plan.py
-
MS_test_update_communication_manager_communication_plan_20260417_170129.log
(FAILED)
-
MA_MS_test_update_communication_manager_communication_plan_20260417_170129.log
(FAILED)
-
MS_test_update_communication_manager_communication_plan_20260417_170210.log
(PASSED)
-
MA_MS_test_update_communication_manager_communication_plan_20260417_170210.log
(PASSED)
===== PASSED:51 | FAILED:2 =====

```

If you need to run the Test Harness on an environment that is using localhost instead of podman networks, remove the "-n ask-network -a activemq-container" arguments to use the default stomp port 61613 or specify the port being used in your environment with (e.g., "--stomp-port 5183").

Shutting down ActiveMQ

Run the appropriate stop scripts

```

$ ./scripts/stop-asb --podman
podman-compose version: 1.0.6
['podman', '--version', '']
using podman version: 4.9.3
** excluding: set()
podman stop -t 10 amq-asb
WARN[0010] StopSignal SIGTERM failed to stop container amq-asb in 10 seconds,
resorting to SIGKILL
amq-asb
exit code: 0
podman rm amq-asb
amq-asb
exit code: 0

```

Notes

Start/Stop Test Signal

The test harness supports stimulating the system under test with a start message and a stop message at start/end of a test to allow for the system under test to acknowledge it is ready to send/receive messages. This feature is turned off by default as it requires the system under test to support a custom UCI extension message that is not yet formalized within the A-GRA ICD. This feature is meant to enable testing of sequences where the system under test is expected to send the first message which are otherwise very difficult to orchestrate and achieve passing results on. An alternative might be to increase the `message_timeout_seconds` in `configs/test-config.yaml` to a much higher value such that you can manually prime your system to send the

message after starting up the test harness. We anticipate a more formal solution to this problem will be presented in a future A-GRA to lower the burden on compliance test orchestration.

To enable this feature, you can set `TEST_TRIGGERS_ENABLED` to `TRUE` in both `podman-compose/host.yml` and `podman-compose/network.yml`.

Example log output with this feature enabled (when System Under Test is not acknowledging the signals).

```
compliance-test-harness-container | INFO      root:amq_asb_utils.py:203
Subscribed to /topic/TestHarness_SignalAck<None>
compliance-test-harness-container | INFO      root:amq_asb_utils.py:604
Published message to /topic/TestHarness_StartSignal<None>
compliance-test-harness-container | WARNING
test.amq_asb_utils:amq_asb_utils.py:738 Did not receive TestHarness_SignalAck
within 5.0s
compliance-test-harness-container | INFO      root:amq_asb_utils.py:203
Subscribed to /topic/TestHarness_SignalAck<None>
compliance-test-harness-container | INFO      root:amq_asb_utils.py:604
Published message to /topic/TestHarness_StopSignal<None>
compliance-test-harness-container | WARNING
test.amq_asb_utils:amq_asb_utils.py:738 Did not receive TestHarness_SignalAck
within 5.0s
```

(Note: these are just warnings and the failure to acknowledge the signal will not be evaluated as a test failure)

Vacuous Tests

There are several sequences in the A-GRA 5.0 ICD that are completely one-sided that will result in a passing sequence even if nothing is connected to ActiveMQ besides the Test Harness itself. This sort of sequence test is included in the Test Harness for completeness and component testing purposes rather than compliance evaluation. Below is a summary of expected vacuous passes for each System Under Test:

System Under Test	Expected Vacuous Results
vi	33 failed, 3 passed
ms	49 failed, 4 passed
c2	172 failed, 36 passed
p2p	223 failed, 44 passed
ma-vi	23 failed, 13 passed
ma-ms	20 failed, 33 passed
ma-c2	202 failed, 6 passed

Troubleshooting

Observation: Every test failed

Symptom: Every test failed extremely quickly, even the vacuous tests that should have passed with nothing running.

Likely cause: ActiveMQ isn't running, or the test-harness-runner script wasn't supplied with the correct network and alias arguments or the right STOMP port.

How to resolve: Ensure ActiveMQ is running. If activeMQ is using the non-default STOMP port, specify the port `--stomp-port` when using the test-harness-runner script. If a custom podman network is needed, make sure the network `--network` is specified as well as an alias `--network-alias` when using the test-harness-runner script.

Observation: A previous log file report vanished

Symptom: Log files from old runs are no longer in the test-harness-logs directory.

Likely cause: The test-harness-runner script has moved them into a historical directory called `old`.

How to resolve: Look for the old log files in `test-harness-logs/old`.

Observation: The UUID for the Test Harness changes on every run (as seen in GridTool)

Symptom: The UUIDs reported in GridTool change on every run.

Likely cause: By default the test harness randomly generates a random UUID for each service.

How to resolve: The UUIDS for services can be set to static values in `configs/test-config.yaml`. For instance, adding the block of yaml below forces the descriptive label of the host system to `System2` and UUID to `b39b83ee-d994-5f64-ae73-e62df8ec3c72`. For P2P (ma-ma) tests, the peer system is forced to use a descriptive label of `System3` and UUID of `a1f29948-4d0a-57d0-9278-f55968b2725b`.

```
platform_system_id:
  system_description: "System2"
  system_uuid_name: 2
  system_uuid_override: b39b83ee-d994-5f64-ae73-e62df8ec3c72
peer_system_id:
  system_description: "System3"
  system_uuid_name: 3
```

```
system_uuid_override: a1f29948-4d0a-57d0-9278-f55968b2725b
```

Observation: The Test Harness did not provide enough time for the System Under Test to respond

Symptom: The Test Harness fails my System Under Test after only 6 seconds, and my system hasn't had the time to compute its response.

Likely cause: The default timeout is set to 6s.

How to resolve: The A-GRA does not currently dictate how quickly a response needs to be provided but it is necessary to avoid tests hanging indefinitely in failure conditions. If your system needs more time to provide responses, you can increase the timeout in `configs/test-config.yaml`. For instance, the block below (default) sets it to 6 seconds.

```
message_timeout_seconds: 6
```